

CENG 476 - INTRODUCTION TO DEEP LEARNING

Plant Disease Classification Using Deep Learning and Transfer Learning

FINAL PROJECT REPORT

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Abstract

Automated plant-disease recognition can support rapid crop-health assessment, but unusually high benchmark scores must be interpreted with care. This project develops a 38-class PlantVillage classification pipeline in PyTorch and compares a custom convolutional neural network trained from scratch with ImageNet-pretrained ResNet18 and EfficientNet-B0 models. An initial image-level split produced an ensemble accuracy of 99.76%, which appeared suspiciously high. A dedicated audit subsequently identified cross-split exact duplicates, perceptual near-duplicates, and photographs mapped to the same physical leaf. The original 99.76% value is therefore retained only as a historical result that motivated the integrity investigation, not as the final benchmark.

The final benchmark uses an ultra-strict protocol based on the PlantVillage official leaf-aware split. Exact collisions were removed from the training side, mapped physical-leaf overlap was eliminated where metadata were available, and all strict cross-split perceptual pairs at dHash ≤ 4 were conservatively quarantined from train or validation while the 10,709-image official test set remained unchanged. Under this protocol, the custom CNN achieved 84.62% accuracy, ResNet18 achieved 97.66%, EfficientNet-B0 achieved 99.01%, and a validation-selected 50/50 ResNet18-EfficientNet ensemble achieved 99.14%. EfficientNet repeated over three random seeds obtained a mean accuracy of 99.001% with a 0.229 percentage-point standard deviation.

Additional controls examined calibration, bootstrap confidence intervals, train-validation-test gaps, random-label sanity, image-corruption robustness, Grad-CAM behavior, and external generalization. The mapped PlantDoc out-of-domain probe produced only 23.31% accuracy for EfficientNet-B0 and 25.00% for the ensemble. The final conclusion is therefore deliberately narrow: near-99% accuracy is reproducible inside the controlled PlantVillage domain after the audited leakage mechanisms are removed, but the result does not imply equivalent real-world performance and the large external-domain drop demonstrates strong domain dependence.

Keywords: plant disease classification; PlantVillage; data leakage; transfer learning; EfficientNet-B0; ResNet18; out-of-domain evaluation; PlantDoc; calibration; reproducibility

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1. Introduction

Plant diseases can reduce crop quality and yield, and diagnosis from visible symptoms is a natural application for image-based deep learning. PlantVillage provides a large, widely used collection of labeled leaf images and is especially useful for studying model architectures, optimization, transfer learning, and interpretability under controlled imaging conditions. However, controlled datasets can also create evaluation pitfalls: repeated photographs of the same specimen, near-duplicate views, simple backgrounds, and narrow acquisition conditions may make a nominal test split easier than independent real-world imagery.

This project began as a conventional comparison among a compact custom CNN, ResNet18, EfficientNet-B0, and a probability-level ensemble. The first image-level protocol produced very high results, culminating in 99.76% ensemble accuracy. Instead of accepting that number uncritically, the project was extended into a reproducibility and data-integrity investigation. The final work therefore has two goals: obtain strong in-domain classification performance and determine how much confidence should be placed in that performance.

1.1 Objectives and contributions

- Build a reproducible PyTorch pipeline for 38-class PlantVillage classification.
- Compare a custom CNN with ImageNet-pretrained ResNet18 and EfficientNet-B0 under consistent preprocessing and optimization rules.
- Audit the original split for exact duplicates, perceptual near-duplicates, and same-physical-leaf overlap.
- Construct an ultra-strict final split that preserves the official test set while quarantining suspicious train/validation counterparts.
- Select checkpoints and ensemble weights using validation data rather than the test labels or test predictions.
- Quantify calibration, uncertainty, seed sensitivity, robustness, and train-validation-test generalization gaps.
- Perform an external PlantDoc out-of-domain probe to distinguish controlled-domain benchmark accuracy from real-world generalization.

Main scientific contribution

The project does not treat the highest accuracy as the only result. It documents why the first 99.76% value was optimistic, rebuilds the benchmark under stricter integrity controls, and then shows that the corrected near-99% PlantVillage performance is reproducible but highly domain dependent.

2. Related Work and Technical Background

2.1 PlantVillage and controlled-domain classification

The PlantVillage repository was introduced to support machine-learning research for plant-health diagnosis [1]. Mohanty et al. later demonstrated that deep convolutional models can achieve very high accuracy on PlantVillage while also showing that performance may deteriorate when models are evaluated outside the controlled acquisition domain [2]. This makes PlantVillage valuable for algorithmic comparison, but it also requires careful wording: a high held-out score is evidence of same-domain recognition, not a guarantee of field deployment performance.

2.2 Transfer learning

ResNet introduced residual connections that improve optimization in deeper networks [3]. EfficientNet uses compound scaling of depth, width, and resolution and achieves strong accuracy-efficiency trade-offs [4]. In this project, both architectures are initialized with ImageNet weights and fully fine-tuned. Transfer learning is particularly relevant because early convolutional features learned from large natural-image corpora provide useful texture, edge, color, and shape representations before task-specific adaptation.

2.3 Regularization, calibration, and explainability

Batch normalization, dropout, data augmentation, AdamW weight decay, learning-rate scheduling, and validation-based checkpointing are used to reduce overfitting and stabilize training [5-7]. Accuracy alone is insufficient for a high-confidence final claim, so the final evaluation also includes Macro-F1, calibration metrics, bootstrap confidence intervals, repeated-seed experiments, corruption stress tests, and Grad-CAM visualizations [8].

2.4 External domain validation

PlantDoc was introduced as a non-laboratory plant-disease dataset assembled from internet-sourced imagery [9]. Its acquisition conditions differ substantially from the controlled PlantVillage images. In this project, semantically compatible PlantDoc classes are manually mapped to PlantVillage labels and used only as an out-of-domain stress test. The resulting score is not treated as a replacement benchmark because the datasets do not have perfectly identical label definitions or distributions.

3. Dataset, Original Split, and Audit Motivation

3.1 PlantVillage characteristics

The local PlantVillage copy contains 54,305 RGB leaf images across 38 crop-condition classes. Images are transformed to 224 x 224 tensors for model input. The dataset is imbalanced across classes, so Macro-F1 is monitored alongside accuracy to avoid allowing large classes to dominate model selection.

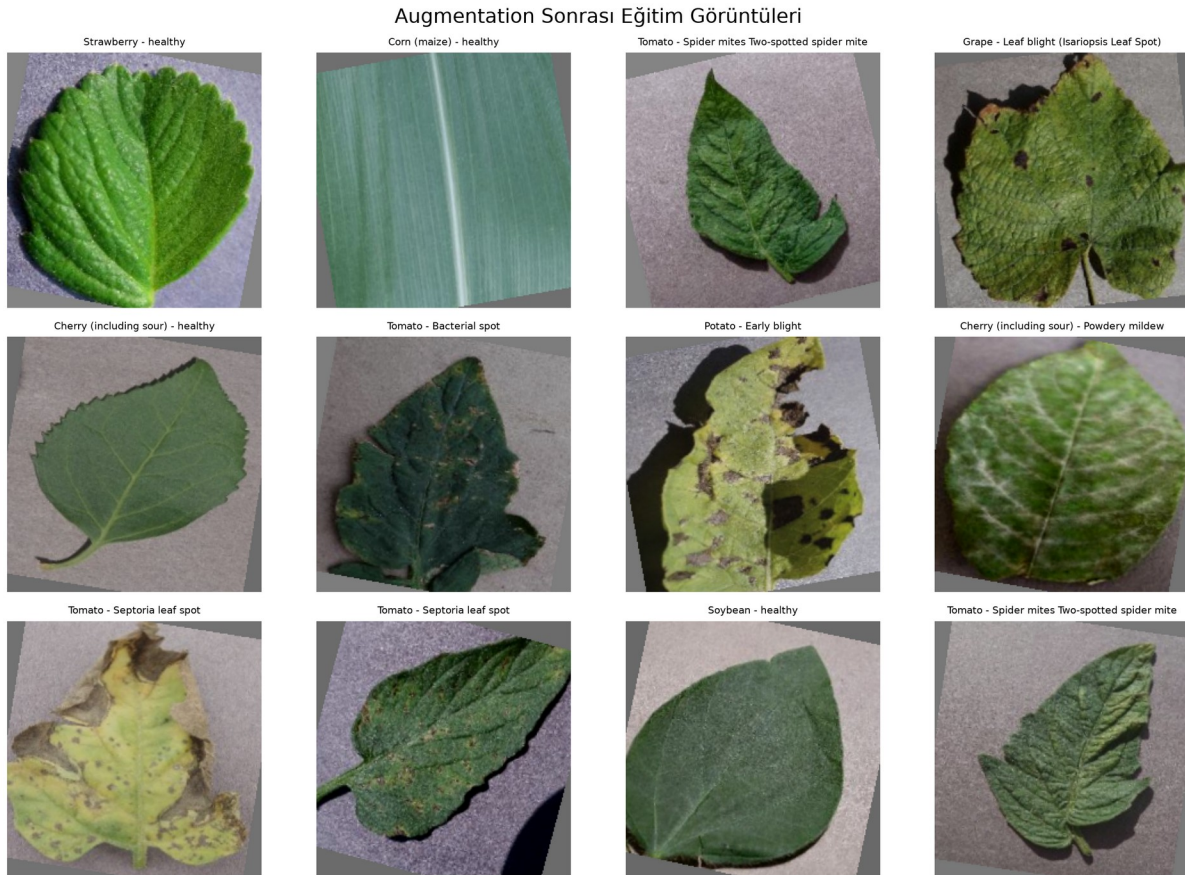


Figure 1. Representative PlantVillage training images after stochastic augmentation.

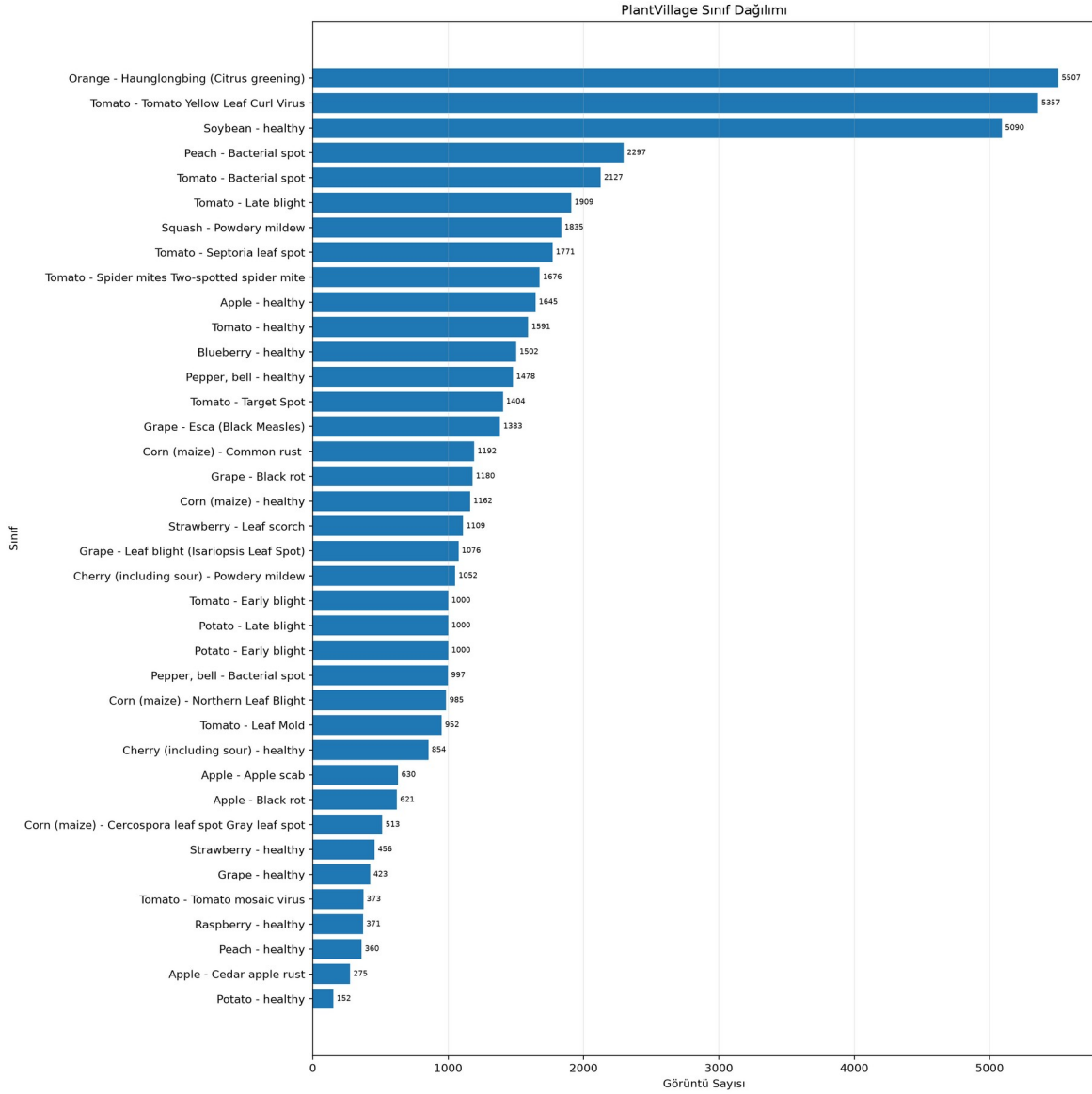


Figure 2. Class distribution in the PlantVillage dataset. The unequal support motivates class-balanced summary metrics such as Macro-F1.

3.2 Original image-level split and suspicious result

The initial development protocol retained 43,444 Kaggle training images and split the supplied validation directory into approximately equal validation and test subsets using stratified image-level sampling. That design ensured index-level separation, but it did not guarantee separation at the physical-leaf or near-duplicate level. The resulting test scores were unusually high, which triggered a dedicated leakage audit rather than immediate final reporting.

Model	Original test accuracy	Interpretation
Baseline CNN	87.42%	Historical development result
ResNet18	99.26%	Historical development result
EfficientNet-B0	99.52%	Historical development result
Ensemble	99.76%	Historical result; not final benchmark

Table 1. Original image-level-split results that motivated the leakage investigation.

3.3 Leakage audit findings

The original split audit identified three distinct overlap mechanisms. First, 10 exact cross-split duplicate pairs were detected by byte/hash comparison. Second, 68 strict perceptual near-duplicate cross-split pairs were detected with dHash ≤ 4 . Third, and more importantly, the available PlantVillage leaf mapping identified 4,956 cross-split groups associated with the same physical leaf. These findings show that the original image-level split was optimistic for specimen-independent evaluation.

- Exact duplicate overlap: detected in the original split.
- Perceptual near-duplicate overlap: detected in the original split.
- Mapped same-physical-leaf overlap: detected in large numbers in the original split.
- Conclusion: the original 99.76% ensemble result is not used as the final benchmark.

The audit does not imply that the entire difference between the original and final scores is caused by one single leakage mechanism. Split composition, stricter data curation, and retraining stochasticity all changed. The correct conclusion is that the original protocol contained leakage risks and was not strong enough for the final claim.

4. Final Ultra-Strict Leakage-Controlled Protocol

4.1 Official leaf-aware split as the base

The final benchmark starts from the PlantVillage official leaf-aware partition rather than the earlier image-level split. The official test portion contains 10,709 images. Five byte-identical train/test collisions were identified in the local mirror and removed from the training side while keeping the official test images unchanged. Where physical-leaf identifiers were available, mapped train/test and train/validation leaf overlap was required to be zero.

4.2 Conservative near-duplicate quarantine

A final strict perceptual audit examined dHash distances across splits. Thirty-nine cross-split pairs met the operational threshold dHash ≤ 4 . Because some unmapped cases visually resembled repeated specimens and leaf metadata cover only about 75.7% of the dataset, the final protocol deliberately used an over-conservative rule: for every strict cross-split pair, the lower-priority training or validation counterpart was quarantined while the test side was never removed. This quarantined 34 training images and 4 validation images and reduced strict dHash ≤ 4 cross-split pairs to zero.

Split	Images	Role
Train	39,091	Optimization and stochastic augmentation
Validation	4,462	Checkpoint, scheduler, and ensemble-weight selection
Locked official test	10,709	Final evaluation and post-hoc diagnostics only
Total used	54,262	38 classes after conservative quarantine

Table 2. Final ultra-strict dataset split.

4.3 Integrity guarantees and boundaries

- Exact train/test hash overlap after cleaning: 0.
- Mapped physical-leaf train/test overlap: 0.
- Mapped physical-leaf train/validation overlap: 0.
- Strict dHash ≤ 4 cross-split pairs after quarantine: 0.
- Test images removed during quarantine: 0.

- Leaf-ID coverage is incomplete, so the protocol does not prove that every unmapped image represents a unique physical leaf.

Test-set usage statement

The locked test set was not used for training, checkpoint selection, hyperparameter tuning, or ensemble-weight selection. Test images were included only in deterministic, model-independent duplicate/near-duplicate integrity auditing and later post-hoc robustness analysis. Test labels and model predictions were not used to choose which training or validation images to quarantine.

5. Models and Optimization

5.1 Custom baseline CNN

The baseline is a compact four-block convolutional network trained from scratch. Each block applies a 3 x 3 convolution, batch normalization, ReLU activation, and max pooling. Channel width increases from 32 to 256, followed by adaptive average pooling, dropout, and a 38-class linear classifier. The final layer returns raw logits because CrossEntropyLoss performs the log-softmax operation internally.

Baseline CNN Architecture (399,142 trainable parameters)

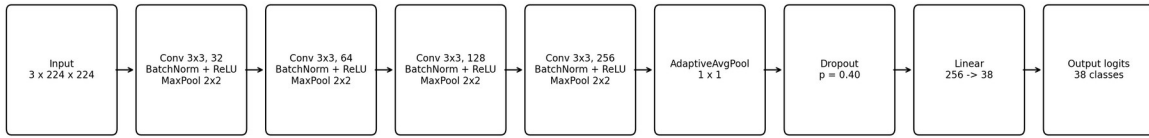


Figure 3. Custom CNN architecture used as the train-from-scratch baseline.

5.2 ResNet18 and EfficientNet-B0 transfer learning

Model	Initialization	Classifier	Parameters
Custom CNN	Random initialization	Dropout(0.40) + Linear(256,38)	399,142
ResNet18	ImageNet pretrained	Dropout(0.30) + Linear(512,38)	11,196,006
EfficientNet-B0	ImageNet pretrained	Dropout(0.30) + Linear(1280,38)	4,056,226

Table 3. Final model configurations.

Both transfer-learning models are fine-tuned end to end. The pretrained backbone uses a lower learning rate than the newly initialized classification head, allowing task-specific adaptation without immediately overwriting the pretrained representation.

5.3 Preprocessing, optimization, and regularization

Training augmentation uses RandomResizedCrop(224, scale=0.80-1.00), horizontal flip with probability 0.50, rotation up to +/-15 degrees, brightness/contrast/saturation jitter of 0.20, and ImageNet normalization. Validation and test preprocessing is deterministic: Resize(256), CenterCrop(224), tensor conversion, and ImageNet normalization.

Setting	Baseline CNN	ResNet18	EfficientNet-B0
Batch size	64	32	32
Maximum epochs	15	12	12

Setting	Baseline CNN	ResNet18	EfficientNet-B0
Backbone LR	5e-4	1e-4	1e-4
Classifier LR	5e-4	5e-4	5e-4
Dropout	0.40	0.30	0.30
Weight decay	1e-4	1e-4	1e-4
Optimizer	AdamW	AdamW	AdamW
Checkpoint metric	Val. Macro-F1	Val. Macro-F1	Val. Macro-F1

Table 4. Final training configuration.

AdamW uses betas (0.9, 0.999) with unweighted cross-entropy. ReduceLROnPlateau monitors validation loss with factor 0.5, patience 2, and minimum learning rate 1e-6. The best checkpoint is selected by validation Macro-F1 with validation loss used as a tie-breaker. Automatic mixed precision is enabled on CUDA.

5.4 Soft-voting ensemble

The ensemble averages class posterior probabilities from the selected ResNet18 and EfficientNet-B0 checkpoints. Candidate weights are compared on validation data only. The 50/50 mixture is selected before the final test evaluation.

ResNet weight	EfficientNet weight	Validation accuracy	Validation Macro-F1	Status
1.00	0.00	98.812%	0.986237	Reference
0.75	0.25	99.059%	0.988722	Candidate
0.50	0.50	99.373%	0.992033	Selected
0.25	0.75	98.969%	0.986400	Candidate
0.00	1.00	98.678%	0.982431	Reference

Table 5. Validation-only ensemble weight search under the ultra-strict protocol.

6. Evaluation and Full-Control Methodology

6.1 Primary test metrics

Final evaluation reports accuracy, macro precision, macro recall, Macro-F1, Weighted-F1, Top-3 accuracy, and one-vs-rest macro ROC-AUC. Macro metrics are emphasized because class support is imbalanced. The official 10,709-image test set is evaluated only after the validation-selected checkpoint is fixed.

6.2 Calibration and uncertainty

Expected Calibration Error (ECE) with 15 confidence bins, negative log-likelihood, and multiclass Brier score are computed on the locked test predictions. A 1,000-resample nonparametric bootstrap estimates 95% confidence intervals for accuracy and Macro-F1.

6.3 Generalization-gap analysis

To investigate conventional overfitting, the selected checkpoint is compared across a deterministic balanced clean-train subset, the validation set, and the locked test set. A severe train-validation or validation-test collapse would raise concern even if the headline test accuracy remained high.

6.4 Random-label sanity control

A balanced subset is trained with shuffled labels while validation retains its true labels. The EfficientNet backbone is frozen and only the classifier is trained. If the pipeline contained a trivial label/path shortcut,

validation performance could remain suspiciously high despite randomized training labels. Instead, performance should remain near the 1/38 chance level.

6.5 Robustness and shortcut stress

Post-hoc image perturbations test sensitivity to brightness, contrast, JPEG compression, Gaussian blur, rotation, and large center/border occlusions. These diagnostics are not used for model tuning. They reveal which image information the trained system depends on and how quickly in-domain accuracy deteriorates under controlled corruption.

6.6 Seed stability and external OOD test

EfficientNet-B0 is retrained on the same fixed ultra-strict manifest with random seeds 42, 123, and 777. All three results are reported together; the highest seed is not promoted as the final benchmark. External generalization is probed with 236 PlantDoc test images from 27 mapped source folders. The models are evaluated without retraining, and the result is interpreted only as domain-shift evidence because PlantDoc and PlantVillage are not identical benchmarks.

7. Results

7.1 Final ultra-strict PlantVillage benchmark

Model	Accuracy	Macro-P	Macro-R	Macro-F1	Weighted-F1	Top-3	Macro AUC	Errors
Baseline CNN	84.62%	0.8635	0.7666	0.7813	0.8361	95.48%	0.995589	1,647
ResNet18	97.66%	0.9734	0.9676	0.9686	0.9763	99.79%	0.999929	251
EfficientNet-B0	99.01%	0.9897	0.9855	0.9874	0.9901	99.91%	0.999955	106
50/50 Ensemble	99.14%	0.9908	0.9891	0.9897	0.9914	99.95%	0.999976	92

Table 6. Final ultra-strict locked-test results on 10,709 PlantVillage images.

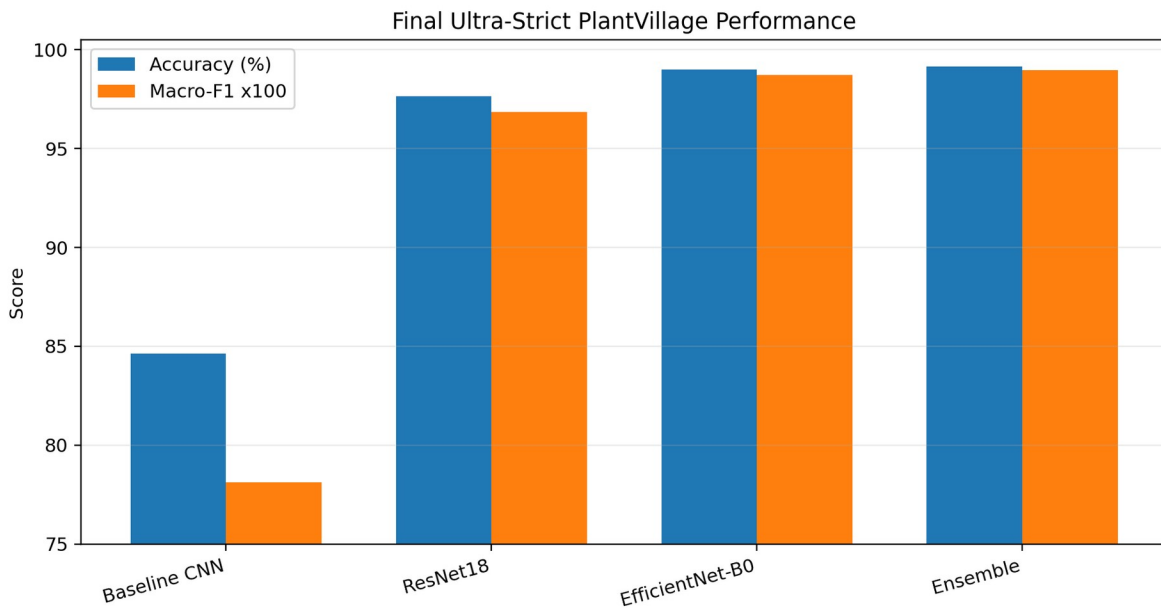


Figure 4. Final in-domain performance after ultra-strict leakage controls.

EfficientNet-B0 is the strongest fixed single-model run with 99.01% accuracy and 0.9874 Macro-F1. The validation-selected ensemble improves accuracy to 99.14% and reduces the error count from 106 to 92. The

custom CNN remains much weaker, demonstrating the value of transfer learning for this controlled-domain classification task.

7.2 Calibration and bootstrap confidence intervals

Model	ECE (15 bins)	NLL	Multiclass Brier	95% accuracy CI
EfficientNet-B0	0.003845	0.035704	0.016145	98.81% - 99.20%
50/50 Ensemble	0.009121	0.035419	0.017435	98.95% - 99.31%

Table 7. Calibration metrics and bootstrap uncertainty.

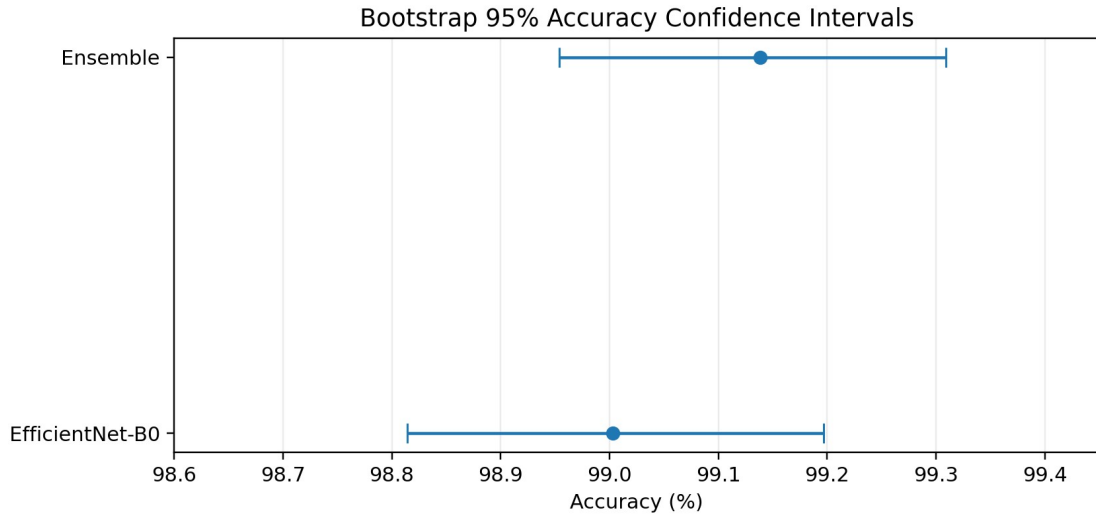


Figure 5. Bootstrap 95% accuracy confidence intervals for the fixed EfficientNet and ensemble predictions.

Both systems are well calibrated on the controlled test distribution, with ECE below 0.01. The bootstrap intervals remain tightly centered near 99%, indicating that the headline result is not an artifact of a very small test set or a single point estimate.

7.3 Generalization-gap analysis

Model	Clean train	Validation	Test	Validation - test
Baseline CNN	77.89%	83.55%	84.62%	-1.07 pp
ResNet18	98.16%	98.81%	97.66%	+1.16 pp
EfficientNet-B0	99.74%	98.68%	99.01%	-0.33 pp

Table 8. Accuracy at the selected checkpoints across clean train, validation, and test data.

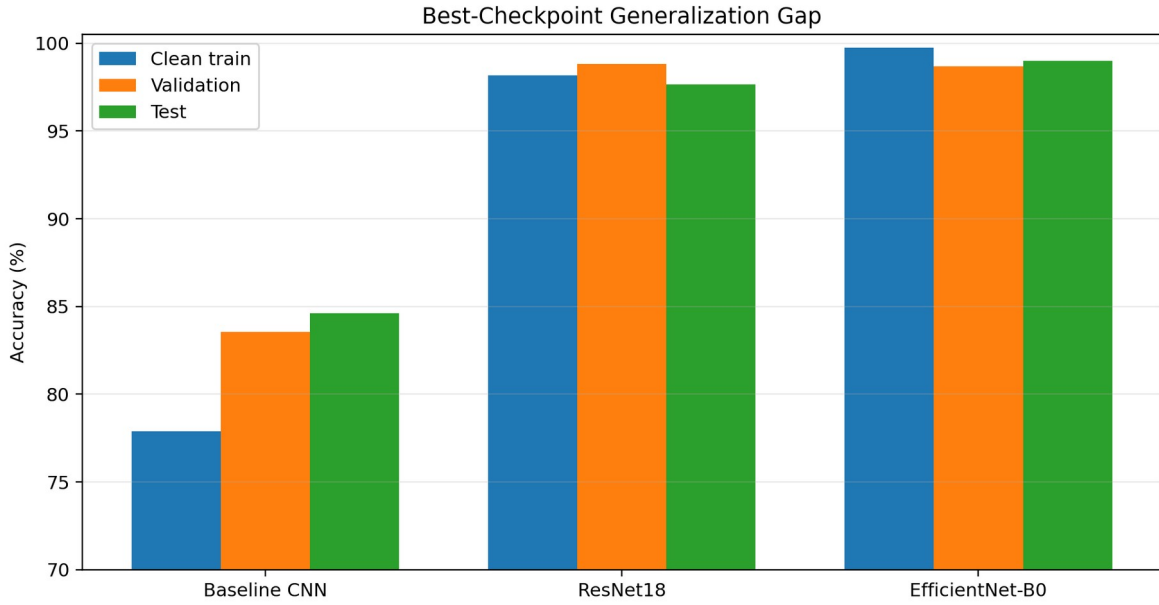


Figure 6. Best-checkpoint generalization comparison.

The transfer models do not exhibit the large validation/test collapse expected from severe conventional overfitting. EfficientNet differs by roughly one percentage point between clean-train and validation accuracy and its test score lies between them. ResNet18 has a modest 1.16 percentage-point validation-to-test drop. These observations do not prove the absence of overfitting, but they do not support severe train memorization as the main explanation for the final near-99% result.

7.4 Random-label sanity

Quantity	Result
Classes / chance accuracy	38 classes / 2.63%
True-label validation accuracy after shuffled-label training	1.84%
Validation Macro-F1	0.0183
Sanity verdict	PASS

Table 9. Random-label pipeline sanity control.

Validation accuracy remains at chance level when training labels are randomized. This is evidence against an accidental direct label/path shortcut in the tested training pipeline. The experiment is a sanity check rather than a mathematical proof that every possible leakage mechanism is absent.

7.5 Robustness stress

Condition	EfficientNet-B0	50/50 Ensemble
Clean	99.01%	99.14%
Brightness x0.60	98.83%	99.18%
Brightness x1.40	98.07%	98.73%
Contrast x0.60	98.49%	98.79%
JPEG quality 30	98.13%	98.72%
Gaussian blur radius 2	84.08%	86.53%

Condition	EfficientNet-B0	50/50 Ensemble
Rotation 15 degrees	99.41%	99.51%
Center occluded 60%	55.38%	64.45%
Keep center 60%, border occluded	58.85%	77.93%

Table 10. Post-hoc corruption and occlusion stress results.

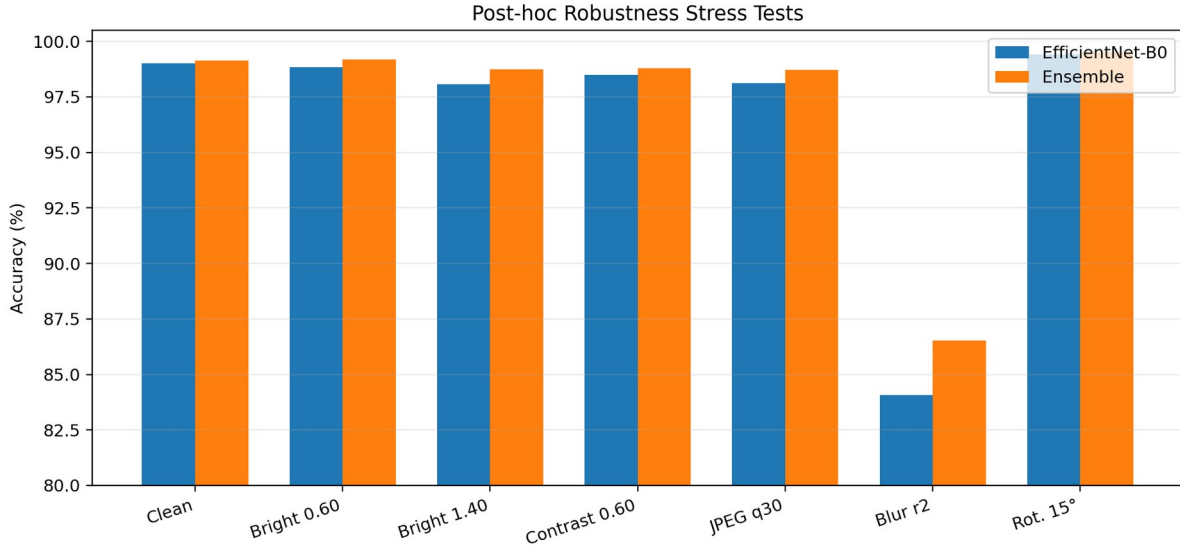


Figure 7. Selected corruption robustness results. Blur produces the largest non-occlusion degradation.

Brightness, contrast, JPEG compression, and moderate rotation have limited effect on the controlled-domain models. Gaussian blur produces a much larger drop, while large occlusions reduce accuracy dramatically. This pattern is consistent with strong dependence on fine-grained visible leaf texture and lesion structure. Occlusion and Grad-CAM are supporting diagnostics only; neither establishes causal feature use.

7.6 Seed stability

Seed	Selected epoch	Validation Macro-F1	Test accuracy	Test Macro-F1	Errors
42	5	0.982431	99.010%	0.987373	106
123	12	0.986704	98.767%	0.983668	132
777	12	0.992056	99.225%	0.990149	83

Table 11. EfficientNet-B0 repeated-seed test results on the same fixed ultra-strict split.

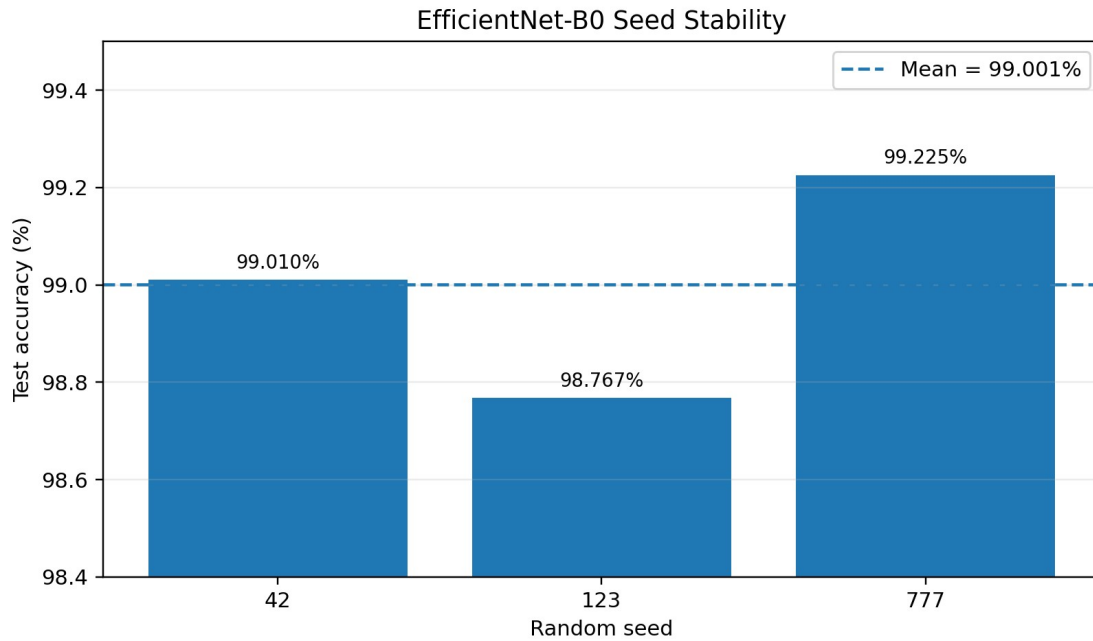


Figure 8. EfficientNet-B0 test accuracy across seeds 42, 123, and 777.

The mean test accuracy is 99.001%, the standard deviation is 0.229 percentage points, and the total range is 0.458 percentage points. The seed-777 run is not treated as the final benchmark simply because it is the highest. Reporting all three runs prevents cherry-picking and shows that the near-99% PlantVillage result is reproducible across stochastic training variation.

7.7 External PlantDoc out-of-domain probe

Model	Mapped PlantDoc images	Accuracy	Mapped Macro-F1	Errors
EfficientNet-B0	236	23.31%	0.2183	181
50/50 Ensemble	236	25.00%	0.2349	177

Table 12. External PlantDoc mapped out-of-domain results without retraining.

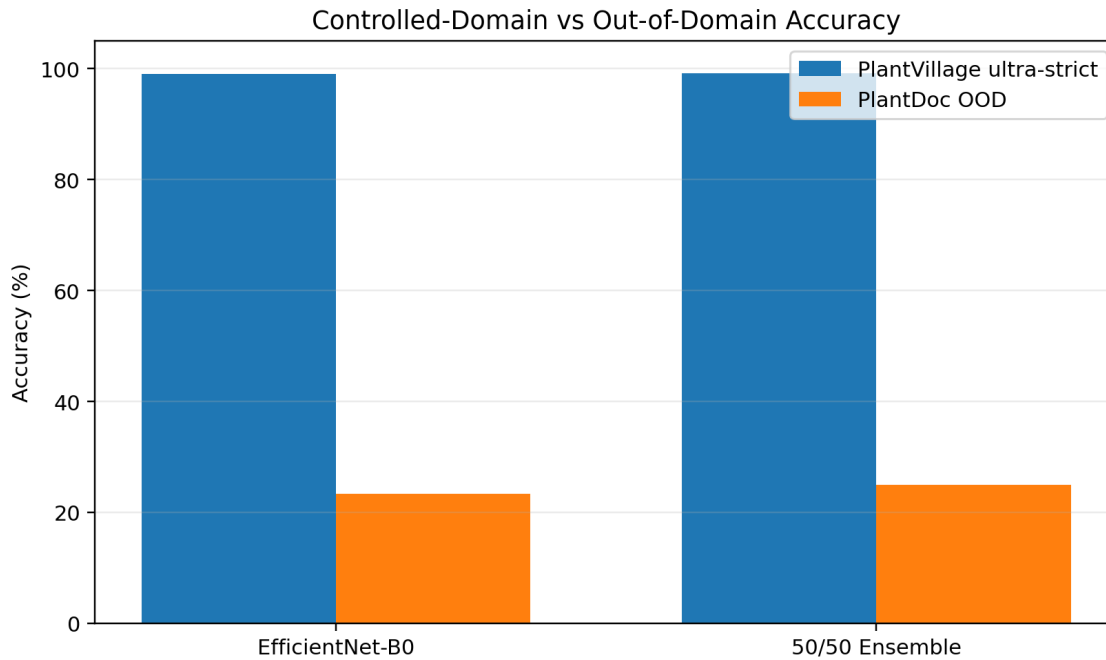


Figure 9. Large accuracy drop from controlled PlantVillage testing to the external PlantDoc domain.

The external result changes the interpretation of the project. The near-99% score is clearly not a universal plant-disease recognition accuracy. PlantDoc imagery differs in background, framing, scale, lighting, disease presentation, and class-definition details. The 23-25% result is therefore not directly comparable to the PlantVillage benchmark, but the magnitude of the drop is strong evidence of domain dependence and limited real-world generalization.

8. Discussion

8.1 What explains the high PlantVillage score?

The final evidence supports a more precise explanation than either 'the 99% score is fake' or 'the model generalizes at 99% everywhere.' The first protocol did contain leakage risks and its 99.76% ensemble result was optimistic. After those audited overlap mechanisms were removed, the fixed ensemble still achieved 99.14% and EfficientNet remained near 99% across three seeds. Calibration and train-validation-test gaps also remained stable. The high controlled-domain result is therefore not explained by the audited leakage mechanisms alone.

At the same time, PlantVillage is visually regular and controlled. Strong transfer-learning models can learn the dataset distribution extremely well. The external PlantDoc result shows that this success is tightly linked to the training domain. In practical terms, the model is a strong PlantVillage classifier, not yet a validated field diagnostic system.

8.2 Conventional overfitting versus domain overfitting

Conventional overfitting usually appears as a widening train-validation gap or a large collapse on a held-out same-domain test set. The final EfficientNet and ResNet checkpoints do not show that pattern strongly. A different phenomenon is more evident: domain over-specialization. The learned representation performs extremely well on the same acquisition distribution but transfers poorly to substantially different imagery. This distinction is central to the final interpretation.

8.3 Why the ensemble is reported separately

The ensemble produces the best aggregate PlantVillage score, but it combines two networks and therefore increases inference cost and operational complexity. EfficientNet-B0 is reported as the best fixed single-model benchmark, while the 50/50 ensemble is reported as an additional validation-selected system. This distinction avoids implying that the ensemble is automatically the preferred deployment model.

9. Limitations and Threats to Validity

Controlled imaging domain. PlantVillage contains largely isolated leaves and relatively simple backgrounds. Same-domain test performance should not be interpreted as field accuracy.

Incomplete physical-leaf metadata. Leaf identifiers cover only about three quarters of the local dataset. The audit can guarantee zero mapped-leaf overlap where identifiers exist, but cannot mathematically prove specimen independence for every unmapped image.

Operational near-duplicate threshold. $dHash \leq 4$ is a deliberately strict engineering threshold. It catches very similar images but is not a complete definition of semantic or specimen-level duplication.

Test-aware integrity auditing. Test images were used in deterministic duplicate and near-duplicate integrity checks, although test labels and model predictions were not used for training or model selection. A reviewer who requires zero test-aware curation could instead construct a group-aware split before any test designation and retrain again.

PlantDoc label mapping. The 236-image PlantDoc evaluation relies on manual semantic mapping between two non-identical label spaces. It is therefore an OOD stress test rather than a directly comparable benchmark.

Finite seed sample. Three training seeds provide useful stability evidence, but they do not fully characterize the distribution of all possible training outcomes.

Robustness diagnostics. Corruption, occlusion, and Grad-CAM tests characterize behavior but do not prove that the model uses only biologically causal features.

10. Conclusion and Future Work

This project evolved from a model-comparison exercise into a controlled investigation of data integrity and generalization. The first image-level split produced an ensemble accuracy of 99.76%, but the subsequent audit detected duplicate, near-duplicate, and same-physical-leaf overlap. That result was therefore retired as the final benchmark. Under the final ultra-strict protocol, the custom CNN achieved 84.62%, ResNet18 achieved 97.66%, EfficientNet-B0 achieved 99.01%, and the validation-selected ensemble achieved 99.14% on the unchanged 10,709-image official test set.

Repeated-seed testing, calibration, bootstrap intervals, random-label sanity, and train-validation-test comparisons do not strongly support severe conventional overfitting as the principal explanation for the final controlled-domain score. The decisive limitation appears when the model is moved outside PlantVillage: accuracy falls to 23.31% for EfficientNet-B0 and 25.00% for the ensemble on the mapped PlantDoc OOD probe. The scientifically defensible conclusion is therefore that the model is highly effective and reproducible within PlantVillage but insufficiently validated for real-field deployment.

Future work should train with more diverse field imagery, evaluate on additional independently collected datasets, investigate domain adaptation and stronger background/lighting augmentation, use pre-split specimen-aware grouping when complete metadata are available, expand repeated-seed experiments, and consider deployment-oriented calibration and abstention strategies for uncertain predictions.

11. Reproducibility

The public repository contains data preparation, model definitions, training scripts, split-integrity audits, ultra-strict manifest generation, full-control diagnostics, OOD evaluation, and repeated-seed utilities. The final full-control workflow is resumable on Windows.

Primary final command

From the project root, run: `call .\run_full_control_all.bat`

11.1 Final artifact paths

- `outputs/audit/official_leaf_safe_split_manifest_ultrastrict.csv` - final train/validation/test manifest.
- `outputs/audit/official_leaf_safe_split_summary_ultrastrict.json` - final split integrity summary.
- `outputs/audit/full_control/FULL_CONTROL_REPORT.md` - consolidated diagnostic report.
- `outputs/audit/full_control/efficientnet_seed_stability_summary.json` - three-seed statistics.
- `outputs/audit/full_control/plantdoc_ood_summary.json` - mapped external-domain result.
- `outputs/evaluation/` - per-model locked-test metrics and predictions.
- `outputs/figures/full_control/` - calibration, error, robustness, and Grad-CAM diagnostics.

11.2 Reporting rule

The report uses 99.01% as the fixed EfficientNet-B0 single-model benchmark and 99.14% as the validation-selected ensemble result. The three-seed mean is reported as a reproducibility statistic rather than replacing the pre-defined fixed run. The historical 99.76% image-level-split result is shown only to document why the leakage audit was necessary.

References

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